

Nam Tùr

MITOCHONDRIA — POWERHOUSE OF THE CELL... AND THE BRAIN TUMOR?

Computational Analysis of Apoptosis and
Mitochondrial Respiratory Complexes in
Classical Glioblastoma Transcriptome

Bachelor's Thesis
Engineering and Natural Sciences
September 2026

ABSTRACT

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The abstract is a concise, self-containing one-page description of the work: what was the problem, what was done, and what are the results. Do not include charts or tables in the abstract. Put the abstract in the primary language of your thesis first and then the translation when that is needed. International students do not need to include an abstract in Finnish.

Keywords: glioblastoma, apoptosis, mitochondria, respiratory complex, transcriptomics

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PREFACE

“DNA neither cares or knows. DNA just is. And we dance to its music.”

— Richard Dawkins

This thesis was written as part of Tampere University’s Bachelor’s Programme in Science and Engineering and was begun in early May 2026. The work was supervised by Dr. Meenakshisundaram Kandhavelu, who inspired my interest in bioinformatics and provided me the opportunity to develop it further in his Molecular Signaling Lab.

This work also marks an important milestone not only for myself but also for my family.

I would like to thank my friends from various parts of the world for their encouragement, insight, and support throughout the writing process — both online and in-person. Finally, I express my deepest gratitude to my parents, V. and K., whose support made this achievement possible. The completion of this thesis — and the degree as a whole — would not have been possible without any of them.

In Tampere on 30. September 2026,
Nam Tūr

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GLOSSARY

CPTAC	Clinical Proteomic Tumor Analysis Consortium, a comprehensive and coordinated effort to understand molecular basis of cancer through the application of robust, quantitative, proteomic technologies and workflows.
DEG	(Significantly) differentially expressed gene.
Apoptosis	A major form of programmed cell death, closely connected to the mitochondria.
Enrichment analysis	Specific method of functional analysis, using statistical methods to identify significant functional groups.
Functional analysis	The identification of biological functions represented by genes or proteins.
<i>EGFR</i>	Italic words denote genes.
GBM	Glioblastoma multiforme, the most aggressive form of brain tumors.
OXPHOS	Oxidative phosphorylation, the driver of mitochondrial respiration.

USE OF ARTIFICIAL INTELLIGENCE IN THIS WORK

Artificial intelligence (AI) has been used in generating this work:

☒ Yes

☐ No

I hereby declare, that the AI-based applications used in generating this work are as follows:

Application	Version
ChatGPT	GPT-5.5

PURPOSE OF THE USE OF AI

Explain here *in detail*, for which purpose and how AI was utilized in writing this thesis.

ChatGPT was used mainly for debugging code during analysis of multiomics.

PARTS OF THIS WORK, WHERE AI WAS USED

List here all chapters, sections, subsections, tables, figures and so forth, that were generated by an AI, or that an AI had a hand in generating.

ACKNOWLEDGEMENT OF RISKS

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1 INTRODUCTION

Glioblastoma multiforme (GBM) is the most aggressive primary brain tumour in adults and remains associated with poor patient survival despite advances in treatment [1]. A major challenge in understanding and treating GBM is its substantial molecular heterogeneity. Molecular analyses have identified distinct GBM subtypes characterised by different transcriptional profiles and biological properties, which can influence tumour progression, treatment response, and patient prognosis [2]. Understanding these molecular differences is therefore important for characterising the biological processes associated with GBM.

Advances in high-throughput transcriptomic technologies have enabled comprehensive characterisation of gene expression patterns in cancer. Transcriptomic data can be used to identify genes whose expression differs between tumour and healthy tissue and to investigate the biological processes associated with these changes [3]. However, lists of differentially expressed genes alone provide limited information about the biological processes represented by the observed transcriptional changes. Functional enrichment analysis can therefore be used to identify biological processes that are disproportionately represented among differentially expressed genes, providing biological context for transcriptomic alterations [4]. These technologies give insight into understanding the behavior and alterations biological processes behind cancer, including that of cell death.

Cell death is an important biological process in cancer, and dysregulation of cell death mechanisms can contribute to tumour development and progression [5], [6]. Among these mechanisms, apoptosis is a highly regulated form of cell death that plays an important role in maintaining cellular homeostasis and eliminating damaged cells. Its intrinsic pathway is closely associated with mitochondrial function, linking apoptosis to mitochondrial processes such as oxidative phosphorylation and the activity of mitochondrial respiratory complexes [5]. Disruption of apoptotic signalling can contribute to uncontrolled cell survival, making the relationship between apoptosis and mitochondrial function relevant to the study of cancer.

This thesis investigates transcriptomic alterations in glioblastoma using data from the Clinical Proteomic Tumor Analysis Consortium (CPTAC) GBM Discovery Study. Differential expression analysis is performed to identify genes whose expression differs between GBM and healthy brain tissue. Gene Ontology enrichment analysis is then used to characterise the biological processes associated with the significant genes, with particular attention given to cell-death-related processes. As apoptosis represents a prominent category among the identified cell-death-related processes, apoptosis-associated genes are subsequently investigated in relation to the mitochondrial respiratory complexes.

Protein–protein association network analysis using STRING is used to examine associations between apoptosis-associated proteins and proteins belonging to the mitochondrial respiratory complexes.

The analysis focuses on the Classical molecular subtype of GBM due to its unique molecular alterations compared to other subtypes [2], enabling the relationship between apoptosis and mitochondrial respiratory complexes to be examined within a defined transcriptional context. The aim of this thesis is therefore to characterise transcriptomic alterations in glioblastoma and investigate the transcriptional and network-level associations between apoptosis and mitochondrial respiratory complexes in Classical GBM through differential expression, functional enrichment, and protein–protein association network analysis.

Chapter 1 has introduced the biological entities of interest, the information to be extracted from them, and methods to be applied. Chapter 2 presents the relevant theoretical background, including glioblastoma, its molecular characteristics, apoptosis, mitochondrial function and respiratory complexes, and computational approaches for transcriptomic analysis. Chapter 3 describes the processing and analysis of the gene expression data, enrichment of the resulting significant gene set, and integration of cell-death- and mitochondrial respiratory-complex-related genes. Results from the analyses are presented in Chapter 4. An in-depth discussion of the results, limitations, and possible future work is presented in Chapter 5.

2 THEORETICAL BACKGROUND

2.1 GLIOBLASTOMA

Glioblastoma multiforme (GBM) is one of the most aggressive and most common form of gliomas, with an extremely poor prognosis of a median survival period of 12-15 months [1], [7]. To date, common carcinogenetic causes of other cancers such as smoking, diet, pesticide exposure, etc. have not been shown to cause GBM [1].

The World Health Organization classifies gliomas into grades I to IV, with grade I being relatively unproliferative and can be eliminated via surgery, and grade IV being the most invasive and highly malignant - GBM is classified under grade IV [1]. Additionally, GBM possesses an extreme heterogeneity in its structure - meaning the tumor is not made up of identical cells, but rather a highly complex mix of different cell types, vast genetic mutations, and biological behaviors - which results in its resistance to conventional therapies and high recurrence rate [8]. A recent review article by David Eisenbarth and Y. Alan Wang [9] identified at least 10 major areas of heterogeneity in GBM's biology, presenting vast amounts of areas to research and treat therapeutically.

2.2 MOLECULAR MECHANISMS OF GLIOBLASTOMA

2.2.1 MOLECULAR HETEROGENEITY AND SUBTYPES

GBM possesses an incredible molecular heterogeneity. Defining transcriptomic mutations of GBM include (i) **TP53 mutations** appearing in 31-38% of all GBM patients which play a role in disrupting cell cycle regulation and apoptosis, (ii) **PTEN mutations** appearing in 24-37% of GBM patients which activate pathways that enduce the survivability of tumors, (iii) **EGFR amplification and mutations** appearing in 36-60% of GBM patients which enhance cell growth, and many others [8], [10]. The expression levels of the genes listed vary vastly on a per-sample basis [1], [2], [9]. This shows that GBM is not a single uniform disease, and different molecular profiles can correspond to different biological characteristics of the same disease.

According to Verhaak et al. and other further sequencing experiments, GBM is categorized into 3 major subtypes according to their expression of signature genes: Proneural, Mesenchymal, and Classical [2], [9]. However, GBM samples have been shown to convert from one subtype to another during tumor progression and recurrence, contributing to its aggressiveness and resistance to therapies. Each subtype have their own distinct levels of expression of the the defining genes and resistance to therapies [9].

The Proneural subtype is characterized by major alteration of *PDGFRA* and point mutation of *IDH1*, and it also had the most *TP53* mutations in the TCGA dataset; functional

analysis suggests the genes are involved in developmental neuron-related processes. Mesenchymal subtype has genes in the tumor necrosis factor super family such as NF1 highly expressed, reflecting its tendency to have higher overall necrosis compared to other subtypes [2], [8].

Verhaak et al. noted that, while *EGFR* are likely important across all subtypes, analysis shows that it has an even more impactful role in the Classical subtype where high-level *EGFR* amplification occurred in 97% of the samples and infrequently occurring in other subtypes. Alongside that, there is a distinct lack of *TP53* mutation in the Classical subtype even though it is the most frequently mutated gene in GBM as a whole [2]. Homozygous deletion of chromosome 9p21.3 also occurred in 94% of the samples, affecting *CDKN2A*. This shows that the Classical subtype is characterized by a distinct molecular profile with alterations in regulatory pathways involving receptor signalling, cell-cycle regulation, and tumor suppression, justifying further investigation of this specific subtype [8]. The functional consequences of these alterations are discussed in the following section.

2.2.2 MAJOR MOLECULAR PATHWAYS ALTERED IN GBM

Molecular alterations in GBM described above significantly affect pathways governing fundamental biological cellular processes, and the dysregulation of these processes promotes growth, invasion, and resistance to therapy in GBM [8].

A few of the most common pathways dysregulated (out of the several dozens) are [8], [10], [11]:

- **RTK/EGFR Signaling:** the protein EGFR is a receptor tyrosine kinase whose activation stimulates downstream signalling pathways including PI3K/AKT/mTOR. Amplification or other activating alterations of EGFR can therefore result in persistent signalling that promotes tumour proliferation and survival.
- **PI3K/AKT/mTOR:** A central pathway influencing cell survival and proliferation with *EGFR* as a major upstream activator, the activation of which promotes tumor growth and treatment resistance. The tumor-suppressing gene *PTEN* negatively affects the pathway, meaning the loss of it exacerbates GBM's aggressiveness.
- **p53:** *TP53* being a major component, the pathway governs cell cycle and apoptosis. The dysregulation of this pathway leads to uncontrolled proliferation and impaired cell death.

Collectively, these alterations have the tendency to disrupt signalling pathways involved in cell proliferation, cell-cycle regulation, and cellular survival. Particularly, alterations affecting survival pathways allow abnormal genes to continue multiplying despite cellular damage or other signals that would otherwise signal their death, contributing to GBM's invasiveness. As such, the cellular survival and death mechanisms are important aspects to be inspected further. Among these mechanisms, apoptosis is particularly relevant because it is among one of the most common forms of programmed cell death [5].

2.3 PROGRAMMED CELL DEATHS

Programmed cell death (PCD) governs the proper life-death cycle of cells, and PCD itself is governed by a complex interaction network of specific genes and proteins. There are four fundamental forms of PCD including autophagy, necroptosis, pyroptosis, ferroptosis, and apoptosis. Of the four, apoptosis is the most common form. Additionally, it was the first form to be described and is studied the most [5], [12].

Apoptosis is essential for development and homeostasis under normal physiological conditions, while its dysregulation has been implicated in various diseases, including cancer. In cancer, impaired apoptotic signalling can allow cells carrying genomic abnormalities to evade elimination and continue proliferating, thereby contributing to tumour development and progression. Consequently, the ability of cancer cells to evade apoptosis is considered an important characteristic of tumorigenesis [5], [11].

Apoptosis has two main pathways depending on the molecular interactions involved: (i) the **intrinsic or mitochondrial pathway**, triggered by a variety of both extra- and intra-cellular stress, and (ii) the **extrinsic or death receptor pathway**, activated when extracellular ligands attach to certain extracellular receptors and resulting in the formation of death-inducing signalling complexes [5], [13]. More specifically, the cascades of interactions leading to apoptosis are described and visualized in Figure 2.1. The main process involves loss of inner mitochondrial membrane potential, hyperproduction of superoxide ions, outflow of matrix calcium glutathione, and release of membrane proteins, resulting in the death of the cell [13].

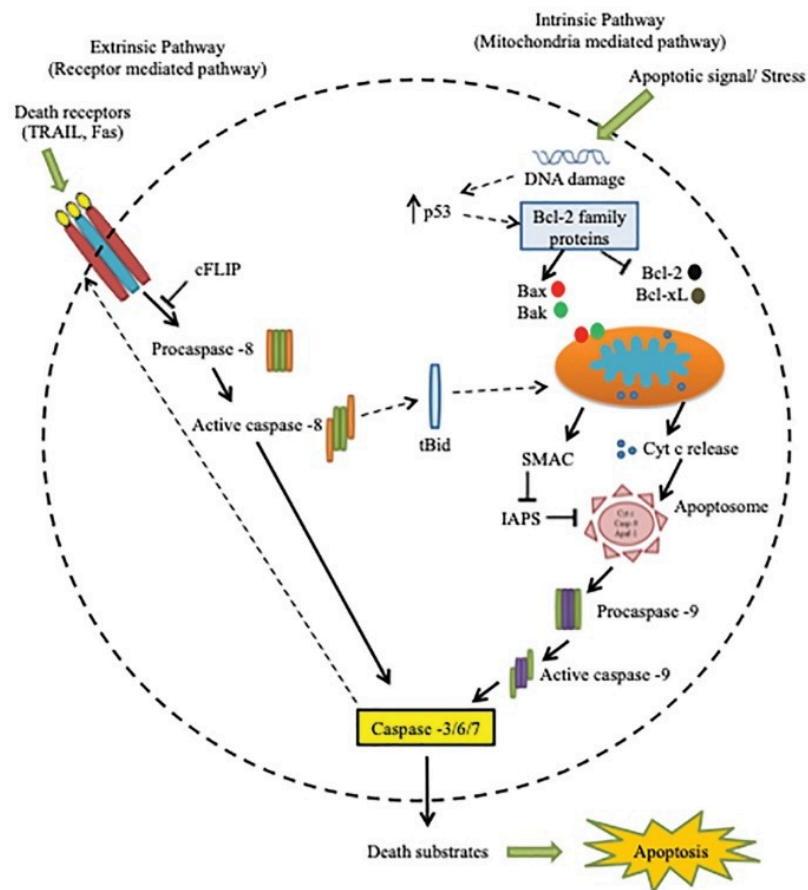


Figure 2.1. Pathways of apoptosis [13]. In the **intrinsic pathway (right)**, stress induces the production of proteins in the Bcl-2 family which when inserted into the mitochondrial membrane, releasing cytochrome c. Cytochrome c interacts with Apaf-1 and procaspase-9, producing the apoptosome complex that triggers the activation of caspase-9. In the **extrinsic pathway (left)** extracellular ligands include TNF (tumor necrosis factor), Fas-L (Fas ligand), and TRAIL (TNF-related apoptosis-inducing ligand), when attached to their respective extracellular portion of transmembrane receptors, forms a death-inducing signalling complex (DISC) and leads to the activation of caspase-8. Both pathways' final caspase ultimately activate executioner caspases which start the main process of apoptosis [13].

Between the two apoptotic pathways, the intrinsic apoptotic pathway is closely associated with mitochondrial function. Mitochondrial outer membrane permeabilization and the subsequent release of cytochrome c represent key events in this pathway. At the same time, mitochondria are responsible for essential metabolic processes, including oxidative phosphorylation through the mitochondrial respiratory chain [5], [13]. This connection between mitochondrial function and apoptosis provides a basis for examining the relationship between apoptosis-associated genes and mitochondrial respiratory complexes, which is discussed further in the next section.

2.4 MITOCHONDRIAL FUNCTION AND RESPIRATORY COMPLEXES

2.4.1 GENERAL MITOCHONDRIAL FUNCTION

The mitochondria is the primary site of cellular function, and it has fundamental roles in processes such as ATP production, ROS generation, and the execution of cell death pathways [13], [14]. Carbohydrates, fatty acids, and amino acids can be metabolised into substrates, which enters the tricarboxylic acid (TCA) cycle. The TCA cycle generates reducing equivalents, primarily NADH and FADH₂, which subsequently donate electrons to the mitochondrial respiratory chain to drive oxidative phosphorylation and ATP production [14].

2.4.2 OXIDATIVE PHOSPHORYLATION AND THE RESPIRATORY CHAIN

Mitochondrial oxidative phosphorylation (OXPHOS) is a key mechanism of energy production in both cancerous and non-cancerous cells. It is comprised of five inner mitochondrial protein complexes I-V (also referred to as CI, CII, etc.) and 2 mobile electron carriers forming the electron transport chain (ETC) [15].

The ETC **oxidizes** reducing equivalents in NADH and succinate, inducing the reduction of molecular oxygen to water and the pumping of protons across the inner mitochondrial membrane (IMM) via Complex I, III, and IV, each using their own pumping mechanism, to drive ATP synthesis [15]. The ETC also uses energy from the reducing equivalents to generate a proton gradient across the IMM, which is used by ATP synthase (Complex V) to **phosphorylate** ADP to ATP, the universal energy currency of cells. More specifically, **Complex I** transfers electrons from NADH to ubiquinone and pumps protons, **Complex II** does not pump protons but contributes to reduction of ubiquinone, **Complex III** reduces ubiquinone and cytochrome c, and in turn shuttles to Complex IV, **Complex IV** donates electron for final reduction of oxygen, and **Complex V** uses the proton gradient and the reduction of oxygen to synthesize ATP [15]. The activity of the respiratory chain is therefore essential for maintaining mitochondrial energy production and cellular function.

2.4.3 MITOCHONDRIAL RESPIRATORY CHAIN-APOPTOSIS RELATIONSHIP IN GBM

The mitochondrial respiratory chain and (intrinsic) apoptosis can be seen to be inherently closely connected to each other. While the respiratory chain is essential for maintaining mitochondrial energy production and homeostasis, intrinsic apoptosis depends on the mitochondria for the regulation of apoptotic signalling. Therefore, disruption of respiratory-chain function and mitochondrial homeostasis can influence apoptotic signalling particularly through cytochrome c because of its involvement in both cellular respiration

and apoptosis (Figure 2.2). This relationship is particularly relevant in GBM, where alterations in mitochondrial metabolism and oxidative phosphorylation have been implicated in tumour-cell survival and progression in specific subtypes [16], [17], [18].

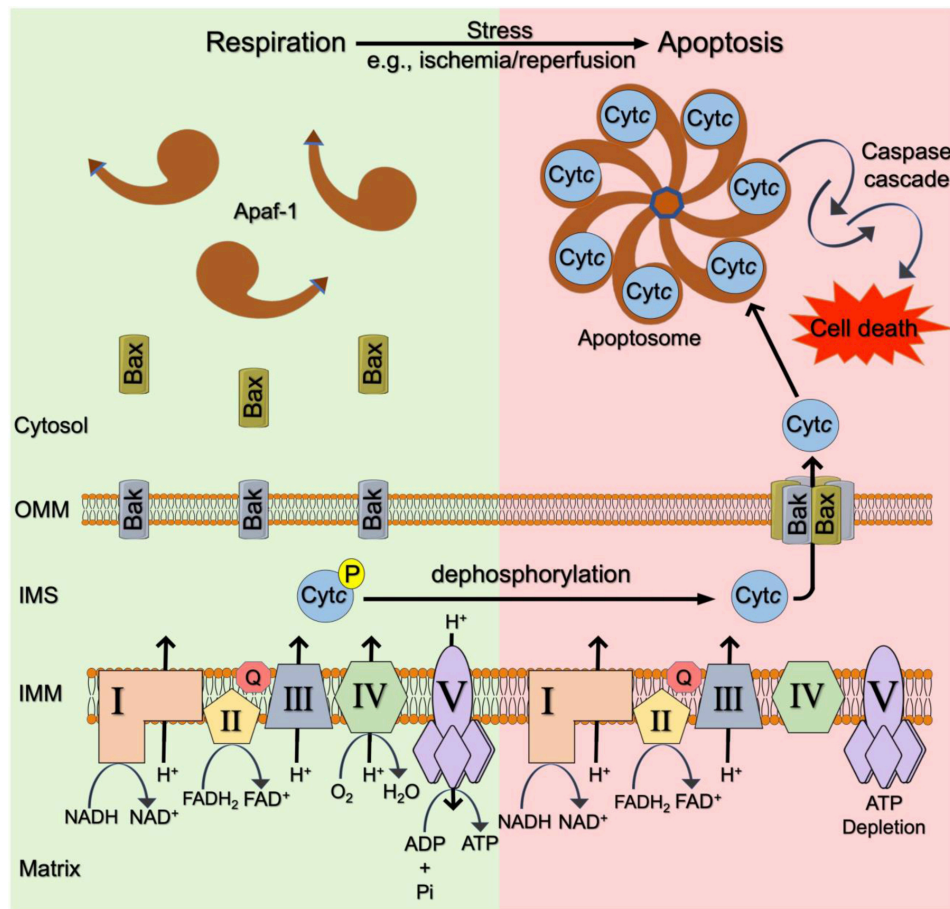


Figure 2.2. Proposed cytochrome c involvement in cellular respiration and apoptosis [18]. Cytochrome c acts as an electron carrier between Complex III and IV. Cellular stress causes cytochrome c to be released from the mitochondria, followed by a variety of interactions that produce apoptosome, resulting in apoptosis [13], [18].

Changes in respiratory-chain activity may therefore influence the susceptibility of GBM cells to apoptosis. More specifically, improper functioning of the mitochondria may result in improper apoptosis, leading to promoted tumor cell survival.

2.5 TRANSCRIPTOMICS AND COMPUTATIONAL ANALYSIS

2.5.1 FROM CONVENTIONAL MOLECULAR ANALYSIS TO COMPUTATIONAL APPROACHES

Having established the relevance of apoptosis and mitochondrial respiratory-chain function in GBM, investigating alterations in the genes involved in these processes requires methods capable of measuring gene activity across large numbers of genes simultaneously - at the systems level.

Understanding complex biological systems requires integration of both experimental and computational approaches [19]. Kitano argues that, although studying individual components of an organism can provide valuable insights, such approaches alone are insufficient to understand the behaviour of the system as a whole [20]. In molecular pathology, diseases are rarely attributable to a single gene, but instead arise from complex interactions between multiple molecular components and biological processes [21].

Although systems-level approaches to biology had already gained considerable interest, their application was constrained by limitations in experimental technologies and the availability of quantitative molecular data [20]. Recent advancements has made high-throughput and high quality profiling of biological systems possible, bringing forth a new set of challenges and opportunities in computationally analyzing and interpreting the new massive amount of deposited data, particularly with RNA-based (transcriptomic) and protein-based (proteomic) data [22], [23].

Among these approaches, transcriptomic technologies provide a means of measuring genome-wide changes in gene expression, which can subsequently be analysed computationally to identify molecular alterations associated with disease. For example, transcriptomic profiling has enabled the identification of distinct molecular subtypes of GBM, presenting more specific areas of deeper research into the disease [2].

2.5.2 TRANSCRIPTOMICS AND DIFFERENTIAL EXPRESSION ANALYSIS

Transcriptomics technologies provide information on gene expression by measuring the RNA transcripts produced from the genome [3]. According to Lowe et al., two major approaches for measuring transcript abundance are microarrays, which quantify predetermined sequences, and RNA sequencing (RNA-seq), which uses high-throughput sequencing to measure transcript sequences without requiring them to be predetermined [3]. Figure 2.3 presents the two technologies and how they quantify expression data in detail.

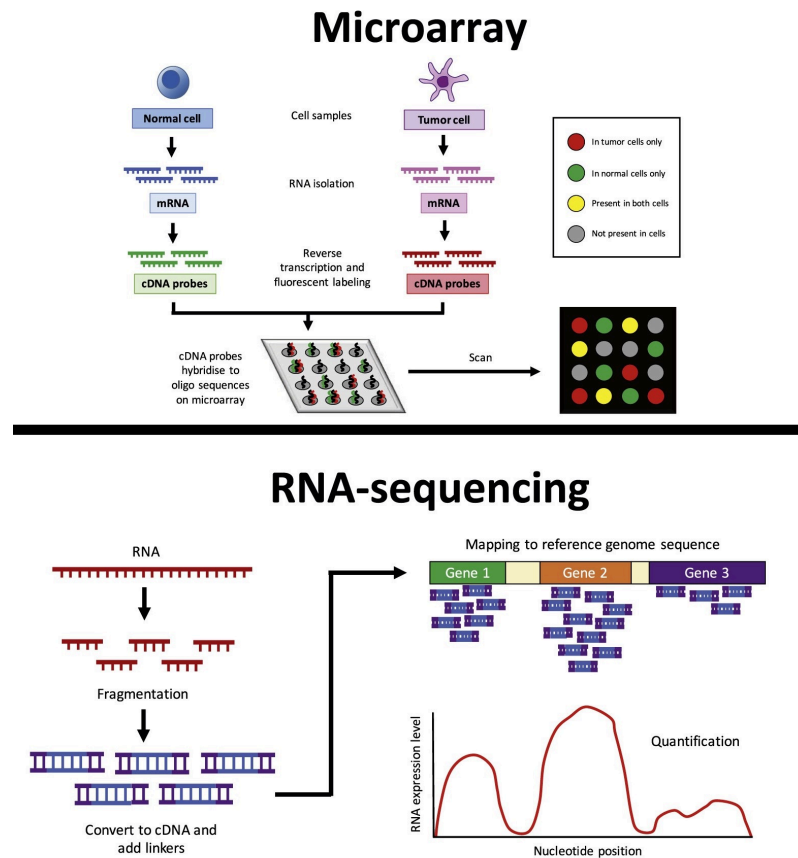


Figure 2.3. Gene expression profiling comparison in microarray and RNA-seq. In microarrays, the extracted RNA are reverse transcribed and labeled with fluorescent probes. The cDNAs are applied to a microarray chip, where they bind to complementary sequences from annotated genes. The expression values correspond to the fluorescence of each cell. In RNA-seq, extracted RNA is fragmented, reverse-transcribed, and modified with linkers to aid sequencing. The obtained cDNA is then sequenced and aligned against a reference genome to obtain the expression levels [24].

Large-scale efforts have been made to profile the transcriptomes of cancers, including GBM. Public resources such as the Gene Expression Omnibus (GEO) and The Cancer Genome Atlas (TCGA) provide access to extensive molecular profiling datasets from numerous cancer types [25], [26]. The Clinical Proteomic Tumor Analysis Consortium (CPTAC) has further generated and integrated transcriptomic and proteomic data from cancer studies, including that of GBM [27]. These resources provide opportunities for researchers to computationally reanalyse previously generated data and investigate molecular characteristics of cancer. The next challenge is therefore to extract meaningful biological information from these large-scale quantitative datasets.

Comparative profiling involves comparing molecular measurements between predefined biological conditions to identify differences associated with a particular phenotype or

state [28]. In transcriptomics, differential expression analysis is commonly used to compare gene expression between conditions, such as diseased and healthy tissue. Its main objective is to identify genes whose expression differs between the conditions, providing insight into potential molecular mechanisms underlying the observed phenotype. The resulting differential expression statistics can describe both the magnitude and direction of an expression difference, commonly represented by the $\log_2$ fold change ($\log_2$ FC), and its statistical significance, represented by a p -value. Thresholds can subsequently be applied to identify genes considered differentially expressed, with the choice of thresholds depending on the experimental context and research objectives [29].

A popular library for gene-level differential expression analysis is DESeq2 (with PyDESeq2 providing a Python implementation) which enables and streamlines vast amounts of analytical possibilities through raw gene counts produced by RNA-seq [30]. An average workflow with DESeq2 suggested by Harvard Chan Bioinformatics Core (HBC) is displayed in Figure 2.4.

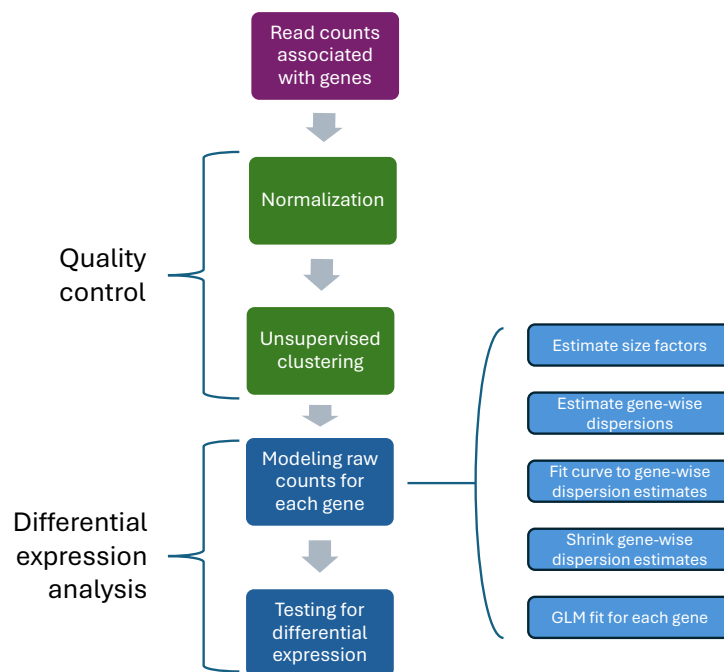


Figure 2.4. Workflow of differential expression analysis using DESeq2 (adapted from HBC). The raw gene count are put through quality control. Afterwards, DESeq2 models raw counts, accounting for size factors, fitting and shrinking gene-wise dispersion estimates, and fitting a generalized linear model for each gene's expression level, producing the gene's corresponding $\log_2$ FC value. The resulting $\log_2$ FCs values are tested for a variety of metrics such as p -values and standard error [31].

RNA-seq count data often exhibit greater variance than their mean, a property known as overdispersion. DESeq2 accounts for this by modelling count data using a negative

binomial generalized linear model, from which $\log_2$ FCs and their standard errors are estimated. The statistical significance of the estimated $\log_2$ FCs is then assessed using Wald tests. DESeq2 can additionally account for known sources of systematic variation, such as batch effects, by incorporating relevant variables into the design of the statistical model. To account for the increased number of false discoveries resulting from multiple hypothesis testing, p -values are adjusted using the Benjamini-Hochberg procedure to control the false discovery rate (FDR) [30], [32], [33].

2.5.3 FUNCTIONAL ANALYSIS AND NETWORK ANALYSIS

Interpreting differentially expressed genes (DEGs) individually may provide limited insight into the broader biological processes underlying the observed changes [20]. Functional enrichment analysis can instead identify biological pathways or functions that are over-represented among the identified DEGs. [34]

A mean of performing functional enrichment is through the Gene Ontology (GO) database. The GO database provides structured and standardized representation of biological activities according to gene products, organized into three main aspects: (1) Molecular Functions (MF), (2) Cellular Components (CC), and (3) Biological Processes (BP) [35]. GO enrichment analysis evaluates whether particular GO terms are statistically overrepresented among a set of genes compared with an appropriate background gene set (Figure 2.5) [34], [36]. The process allows groups of DEGs to be interpreted in terms of shared biological functions and processes rather than as individual genes.

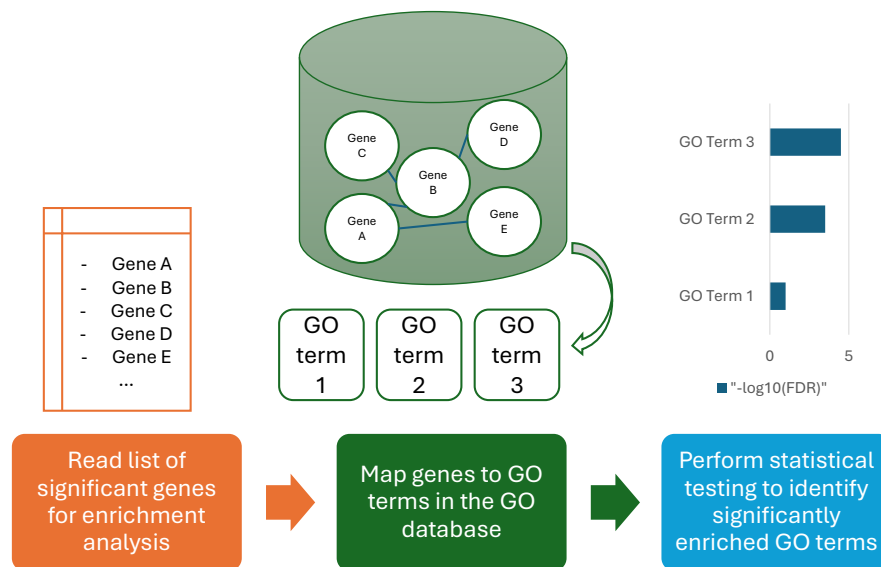


Figure 2.5. Workflow of GO enrichment analysis. The list of DEGs are fed into the GO database which maps the DEGs to specific GO terms. The terms are statistically tested to determine its enrichment level. Adapted from The Gene Ontology Consortium [37].

However, the identified overrepresented biological processes do not show how the individual genes are interacting with one another. Therefore, a network-based approach can complement the identified BPs by representing the genes as nodes and their interactions as edges, allowing for the inspection at the systems-level [21]. Network-based approaches can reveal highly connected components, clusters of interacting molecules, and relationships between biological processes that may not be apparent when genes are considered individually. This can provide additional context for interpreting molecular alterations and identifying potential interactions between genes involved in related biological processes [21].

In this thesis, Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) was used to construct protein-protein association networks from selected differentially expressed genes. STRING maps the submitted gene identifiers to their corresponding protein products and integrates known and predicted protein associations from multiple evidence sources. The resulting networks represent proteins as nodes and their associations as edges, providing a means of examining functional relationships among the proteins encoded by the selected DEGs [38].

A useful metric STRING provides from individual interactions is the STRING combined score which is computed by probabilistically combining multiple independent evidence sources, ranging from 0.150 to 0.999. STRING's 7 evidence channels include text mining, experiments, curated databases, co-expressions, co-occurrence across genomes, gene neighbourhoods, and gene fusion. A closer score to 1 indicates higher confidence that the interaction is real [39].

Together, functional enrichment and network analysis therefore enable differentially expressed genes to be interpreted at a systems level, providing both information on the biological processes associated with gene-expression changes and the molecular relationships connecting the genes involved.

3 MATERIALS AND METHODOLOGY

3.1 DATA SOURCE

The transcriptomic data were retrieved from the publicly available Clinical Proteomic Tumor Analysis Consortium Glioblastoma Multiforme (CPTAC-GBM) cohort through the National Cancer Institute's Genomic Data Commons (GDC) Data Portal. The GDC dataset contained RNA-seq data from over 200 GBM samples in the form of raw counts of each gene. CPTAC-GBM Discovery Study metadata were then used to identify the molecular subtypes of the corresponding cases. Among the CPTAC cases, 99 samples with the Proneural, Mesenchymal, and Classical molecular subtypes were identified and used for the transcriptomic analysis [40], [41]. Of the 99 samples, 25 were identified to be of the Classical subtype.

Although CPTAC provides RNA-seq data for "healthy" samples, the samples are Normal Adjacent Tissue (NAT) which are brain tissues that have yet to exhibit cancerous features nearby the extracted tumors [40], [42]. Aran et al. showed NAT samples specifically from CPTAC's cancer samples have substantial transcriptomic and epigenetic alterations, causing them to be in a molecular "unique intermediate state" between healthy and tumor, despite not yet exhibiting cancerous features [43]. Therefore, NAT samples provided by CPTAC were considered unsuitable as a baseline for comparison in this thesis. In its place, the Genotype-Tissue Expression (GTEx) project provides comprehensive, and peer-reviewed RNA-seq data from healthy brain tissue, suitable for comparison in cancer research [44], [45].

The CPTAC-GBM Discovery Study used 10 normal frontal cortex samples from GTEx as controls [41]. Of these 10 GTEx controls, transcriptomic data for 7 were accessible and were therefore used as controls in this thesis. The same GTEx controls were selected to maintain consistency with the original CPTAC-GBM study and to facilitate potential future comparisons between transcriptomic and proteomic data.

3.2 DATA QUALITY CONTROL AND INSPECTION

As suggested by the Harvard Chan Bioinformatics Core, with raw gene counts, a number of unsupervised exploratory data analysis methods can be used to inspect the general landscape of the data and assess its quality, mainly through hierarchical clustering methods and Principal Component Analysis (PCA) of log2-transformed count data. Because transcriptomic expression values can span several orders of magnitude, log2-transformation compresses the range of expression values and reduces the influence of highly expressed genes, thereby improving the suitability of the data for clustering and visualization [46], [47], [48].

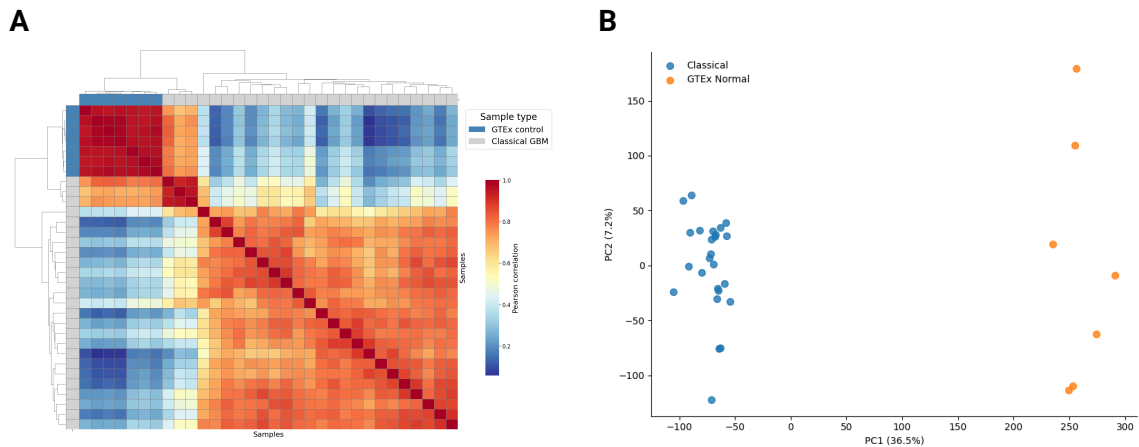


Figure 3.1. Data Quality Control Analyses. **(A)** Hierarchical clustering heatmap of sample-to-sample correlations, demonstrating group-wise similarities. **(B)** Principal Component Analysis (PCA) plot showing the global variance and distinct separation between sample cohorts.

The hierarchical clustering heatmap (shown in Figure 3.1) groups samples according to their similarity based on the samples' gene expression Pearson correlation. Samples with similar biological characteristics are expected to exhibit similar expression profiles and therefore cluster together [48]. In the current dataset, most Classical GBM samples form a distinct cluster from the control samples. However, three samples cluster more closely with the controls than with the other GBM samples. To determine whether these samples represent potential outliers or reflect biological variation, PCA is a complementary quality control plot used for further data quality confirmation.

As noted by the Harvard Chan Bioinformatics Core [48], biological replicates are expected to have similar expression profiles and cluster together in PCA. The PCA (also shown in Figure 3.1) demonstrates clear separation between Classical GBM and control samples, with no apparent sample-level outliers. Thus, although three samples show higher similarity to the controls in the hierarchical clustering analysis, their position in the PCA does not indicate clear evidence of them being outliers.

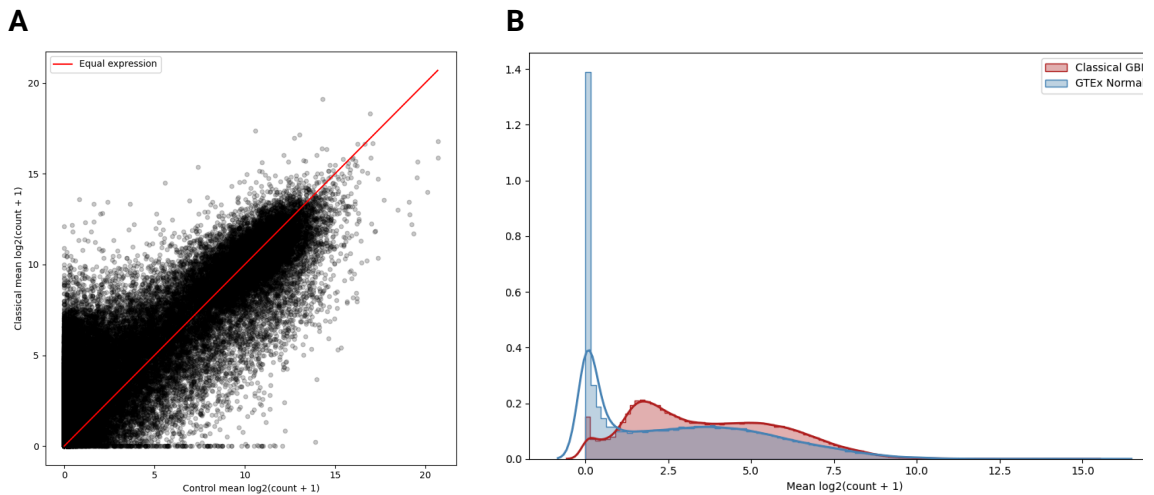


Figure 3.2. Comparative Gene Expression and Variance Distributions. **(A)** Scatter plot of gene expression magnitudes comparing Classical GBM against GTEx control samples. **(B)** Frequency histograms showing the distribution profiles of the top 10,000 genes exhibiting the largest absolute differences in expression.

Figure 3.2 presents an initial exploration of the baseline expression levels across the two cohorts. The scatter plot reveals a subset of genes exhibiting high expression values in the GBM samples relative to the GTEx controls. This is further confirmed by the expression histogram of the 10,000 genes of 45,000 with the largest differences between the GBM samples and controls, where expressions in GBM is generally shifted upward compared to the control. This further validates the need for further investigation into the differences between the two.

3.3 DIFFERENTIAL EXPRESSION ANALYSIS

The raw count data from the 25 Classical GBM samples and 7 GTEx control samples were used as input to PyDESeq2 to obtain the $\log_2$ FC values and their corresponding adjusted p -values. Volcano plots were used to display the distribution of differentially expressed genes (DEGs). DEGs $|\log_2 \text{FC}| > 1$ and $p < 0.05$ were considered significantly differentially expressed. For further detail, significant DEGs with $\log_2 \text{FC} > 1$ were labelled as “up-regulated”, and $\log_2 \text{FC} < -1$ as “down-regulated”. These thresholds were selected to identify genes showing both a minimum two-fold change in expression and statistically significant differential expression. The volcano plot along with the count of significant DEGs are presented in Section 4.1.

3.4 GO FUNCTIONAL ENRICHMENT

GO enrichment analysis was performed on the set of significant differentially expressed genes (DEGs) to identify overrepresented Biological Process (BP) terms. The analysis was performed using the GSEapy Python library with the predefined GO_Biological_Process_2023 gene-set library. The significant DEGs identified from

the differential expression analysis were used as the input gene set. The resulting enrichment terms were ranked according to their statistical significance and the number of genes involved each of the enriched term, and displayed in a dot plot. The results are presented in Section 4.2.

The top enriched biological processes were subsequently examined to identify processes relevant to the objectives of this study. Based on the enrichment results and discussed biological connection in Section 2.4.3, apoptosis and mitochondrial respiratory-chain complexes were selected for further analysis.

3.5 STRING INTERACTION NETWORK MAPPING AND ANALYSIS

The list of genes associated with apoptosis was retrieved from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, via GSEAPy using the `KEGG_2021_Human` gene-set library [49]. The list of genes complexes I–V was retrieved from Human Mitocarta which provides information on mitochondrial genes such as their pathways and tissues in which they are involved [50]. DEGs belonging to either the apoptosis gene set or the mitochondrial respiratory-complex gene sets were combined into a single gene set for further analysis. As an additional visualization, $\log_2FC$ of the genes in the set were plotted in a bar graph to inspect the portion of up and down-regulated genes in each of the biological entities involved.

The selected genes are then fed into to the STRING database to obtain protein association networks, limiting the confidence score of interaction to at least 0.90 to see the most meaningful interactions. The resulting networks were imported into Python and analysed using the NetworkX library.

Associations between apoptosis-associated proteins and proteins belonging to each mitochondrial respiratory-chain complex were identified by counting the corresponding network edges and identifying the proteins involved. The resulting edge counts and protein involvement were used to characterize the associations between apoptosis and complexes I–V. The corresponding results are presented in Section 4.3.

3.6 STATISTICAL ANALYSIS

The differential expression analysis was performed with PyDESeq2. The $\log_2 FC$ values were modelled with a negative binomial generalized linear model, and the corresponding p -values are determined using Wald tests. The p -values were further corrected with Benjamini-Hochberg method to account for inflated false discovery rate due to multiple testing [30], [33]. $\log_2FC > 1$ with $p < 0.05$ were considered significant.

The p -values for GO Enrichment were determined using one-sided Fisher's exact test, and also corrected using Benjamini-Hochberg method for multiple testing; $p < 0.05$ were considered significant [51]. The STRING confidence scores are based on the database's own probabilistic framework. A closer score to 1 indicates higher confidence that the interaction is real [39].

4 RESULTS

4.1 DIFFERENTIALLY EXPRESSED GENES IN GBM

To identify genes that are meaningfully differentially expressed and their distribution, a volcano plot was used. Displayed in Figure 4.1 is the volcano generated from the differential analysis results performed with PyDESeq2, with $\log_2$ FC on the x-axis and its $-\log_{10}(\text{adjusted } p\text{-value})$ on the y-axis for a single gene. The plot shows a large portion of the genes in the GBM samples are significantly differentially expressed, and the change in expression and their statistical significance vary. Absolute values $\log_2$ FC can range from 0 to 15, and $-\log_{10}(p)$ can reach over 300, corresponding to $p = 10^{-300}$.

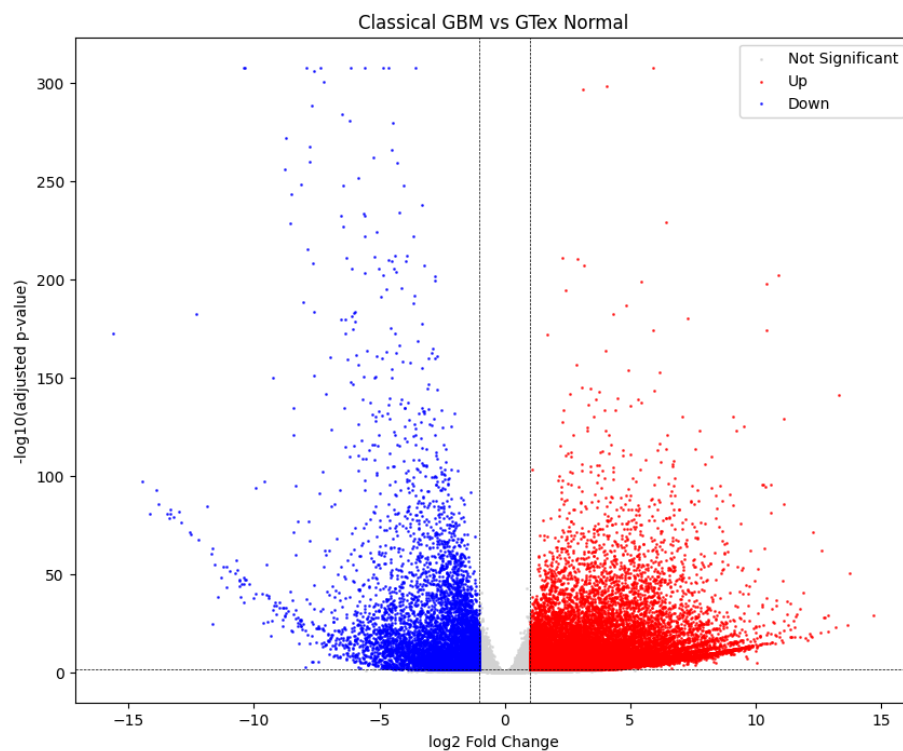


Figure 4.1. Classical GBM vs GTex Control Volcano Plot.

Of the 45,000 results produced by PyDESeq2, roughly 27,300 genes were identified to be significantly differentially expressed - around 60%. More specifically, 20,300 of the 27,300 were up-regulated with mean $\log_2$ FC of 3.98, and the remaining 7,000 genes were down-regulated with mean $\log_2$ FC of -2.71 . The bulk of the $\log_2$ FC values lie within approximately 5 $\log_2$ FC units for downregulated genes and 10 $\log_2$ FC units for upregulated genes.

4.2 FUNCTIONAL ENRICHMENT OF DIFFERENTIALLY EXPRESSED GENES

To identify biological processes (BPs) that may be affected in GBM compared to a healthy sample, functional enrichment is performed to identify BPs that are represented by different subsets of genes within the set of identified DEGs and their statistical significance. With the set of significant DEGs obtained from the differential analysis, they are enriched with GO to identify overrepresented biological processes in the GBM samples. The results of the enrichment are displayed as a dot plot in Figure 4.2 where the top 20 overrepresented GO Biological terms are presented along with their statistical significance and the count of their representing genes.

Biological processes with connections to mitochondrial respiration, either directly stated its name or are a smaller process making up the entire respiration process, are highlighted in red to visualize their portions relative to the other identified BPs. Of the top 20 overrepresented BPs in terms of gene counts, 9 were identified to be related to the respiratory functions of the mitochondria; all with corresponding $p < 0.05$. Furthermore, 7 out of 9 identified mitochondrial respiration-related BPs lie within the top 10 overrepresented BPs.

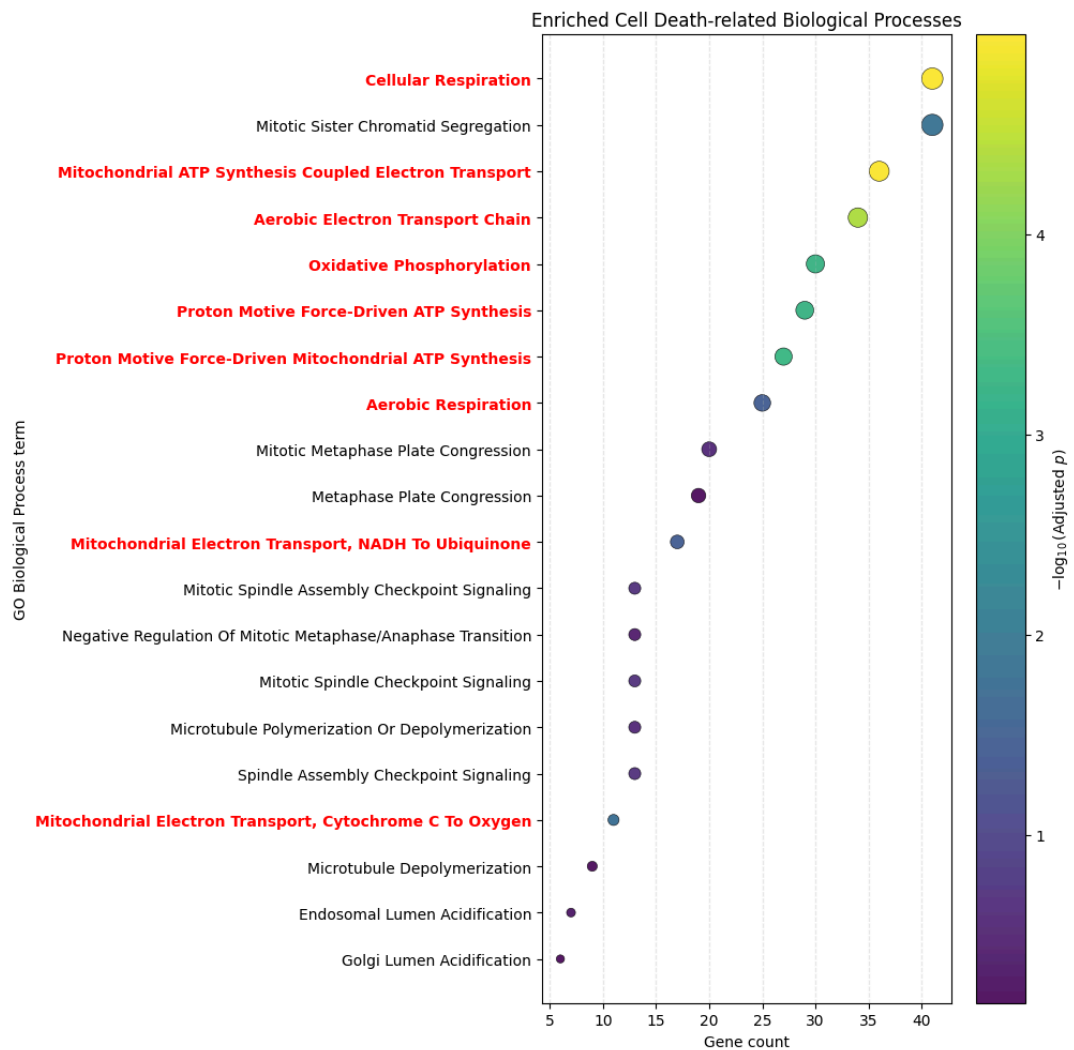


Figure 4.2. GO Enrichment dot plot. Terms were ranked based on count of representing genes. Terms with same number of representing genes are ranked based on $-\log_{10}(\text{adjusted-}p)$. Terms related to mitochondrial respiration are highlighted in red. Of the top 20 overrepresented BPs, 9 were involved in mitochondrial respiration; all with corresponding $p < 0.05$.

This shows that the dysregulation of the mitochondrial respiration process does not occur by chance. Because this process has a crucial role in maintain normal mitochondrial function, the resulting improper apoptosis may also not occur by chance; further validating the lack of cell death being one of the main drivers of GBM proliferation. For further inspection of the genes representing the mitochondrial respiration processes, the genes' $\log_2$ FC magnitudes are visualized in a bar graph.

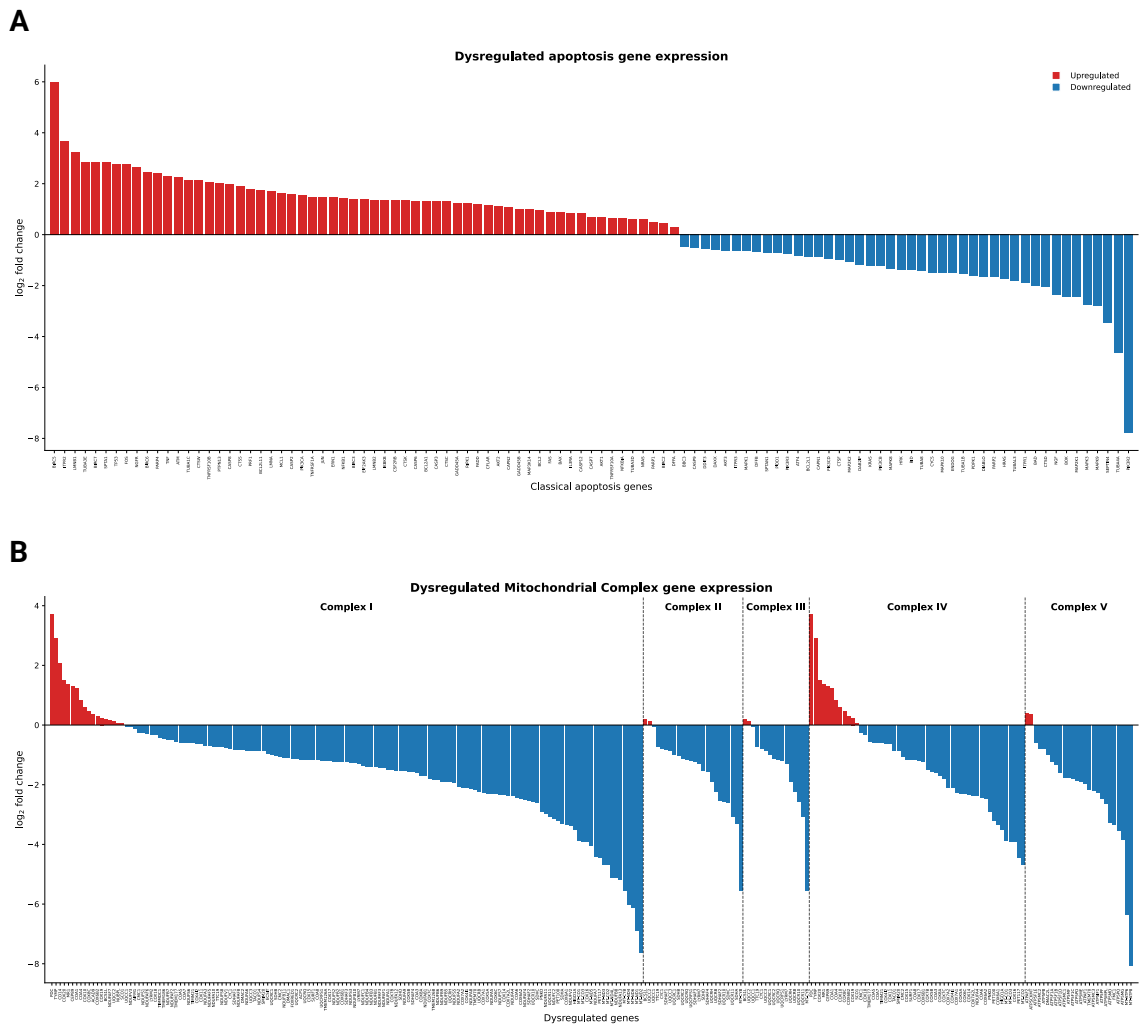


Figure 4.3. Log₂FC bar graph of genes in (A) in apoptosis, and (B) respiratory complexes.

Figure 4.3 shows the log₂ FC magnitudes of the genes involved in respiratory complexes and in apoptosis. Proportion-wise, all the complexes are seen to be mostly down-regulated, while apoptosis is split down the middle between up- and down-regulated. The biological implications of the above results are discussed in Section 5.1.

4.3 ASSOCIATION BETWEEN APOPTOSIS AND MITOCHONDRIAL RESPIRATORY COMPLEXES

Displayed in Figure 4.4 is the obtained interaction network containing proteins resulting from apoptotic and respiratory complex genes. Additionally, the protein's involvement is annotated by the color it is filled.

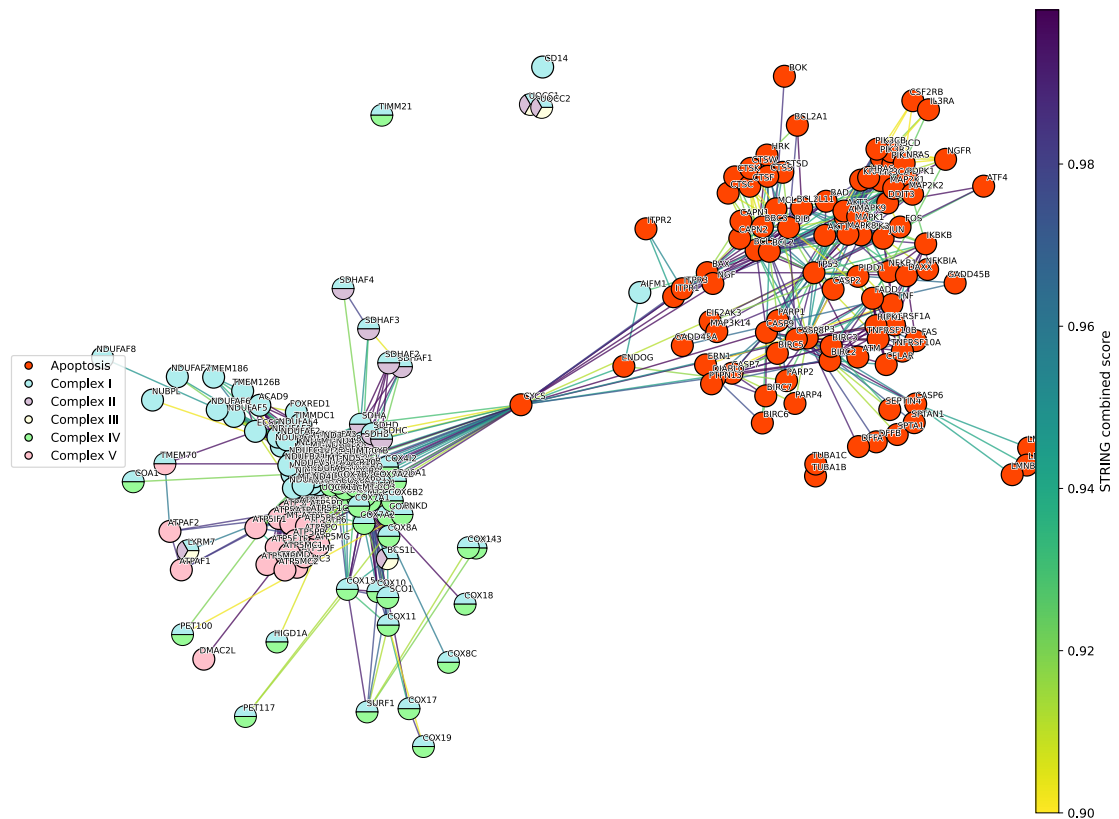


Figure 4.4. Protein interaction network from apoptotic and respiratory complex genes in classical GBM. Each protein (node) is colored with its corresponding involvement(s). The edges are colored according to the STRING confidence scores. CYCS is observed to be a majorly involved protein, connecting the two separated cohorts of respiratory complexes and apoptosis proteins.

The 2 major groups can be seen to be very distinct from one another, and connect with each other through a single apoptotic protein that is CYCS. The STRING combined score of the interactions present start from at least 0.90 to as high as 0.999, with many of the interactions connected to CYCS lie in the higher end of the range. This implies that the gene resulting in CYCS plays a major role in mediating the interaction between the 2 groups.

The relationship between mitochondrial energy metabolism and apoptosis is particularly relevant in GBM, as mitochondrial processes are involved not only in energy production but also in the regulation of apoptotic pathways [16]. In particular, cytochrome c, encoded by identified CYCS protein, has a dual role as a component of the mitochondrial electron transport chain and as a central mediator of the intrinsic apoptotic pathway. Under apoptotic conditions, the release of cytochrome c from mitochondria into the cytosol contributes to the formation of the apoptosome and subsequent activation of downstream caspases [13], [18].

For further inspection, the interactions between apoptosis and each of the respiratory complex are mapped in Figure 4.5, and their log₂ FC values are filled within each node. A major standout in the individual networks is the interaction network between apoptosis and complex V (second subfigure from left to right, second row), where there exists no connection between 2 the groups.

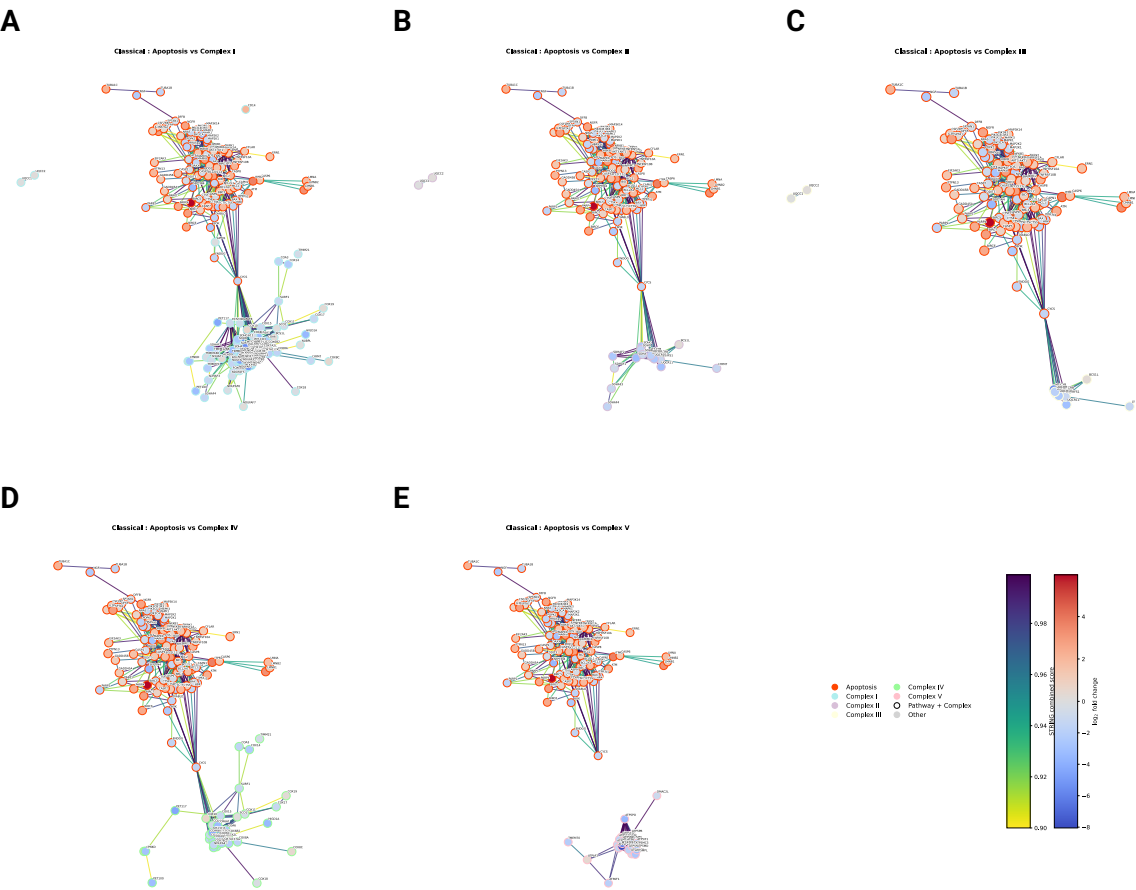


Figure 4.5. Protein interaction subnetworks in Classical GBM. Subnetworks are between (A) apoptosis and complex I, (B) apoptosis and complex II, (C) apoptosis and complex III, (D) apoptosis and complex IV, and (E) apoptosis and complex V.

Beyond the established CYCS-mediated connection between mitochondrial respiration and apoptosis, the subnetworks analysis can identify additional associations between apoptotic genes and the individual mitochondrial respiratory complexes. These findings extend the analysis beyond the canonical relationship and provide a more detailed view of how apoptotic genes may be connected to different components of the mitochondrial respiratory chain in Classical GBM.

Table 4.1. Interaction subnetwork edge count

	Complex I	Complex II	Complex III	Complex IV	Complex V
Apoptosis	25	13	9	10	0

The count of edges connecting apoptotic and respiratory complex genes are shown in Table 4.1. The interactions of each of the complexes with apoptosis can vary. Excluding complex V which had 0 interactions, complex III has the lowest number of interactions of 9, and complex I with the highest number of interactions of 25, followed by complex II with 13, complex IV with 10, and complex III with 9.

5 DISCUSSION

5.1 TRANSCRIPTOMIC ALTERATIONS AND MITOCHONDRIAL PROCESSES IN CLASSICAL GBM

The large proportion of DEGs implies that the transcriptional differences between Classical GBM and normal brain tissue spans across a broad range of biological functions rather than being restricted to a small number of genes. The volcano plot also indicated an asymmetric distribution of log2FC values, with most genes exhibiting moderate changes and a smaller subsets showing a substantially larger fold changes. This further validates the highly heterogeneous profile of GBM as described in Section 2.2.1.

To investigate the biological processes associated with these transcriptional alterations, GO enrichment analysis was performed on the significant DEGs. The enrichment results indicate that genes involved in mitochondrial respiratory processes were disproportionately represented among the genes showing significant transcriptional differences in Classical GBM.

The strong representation of mitochondrial respiratory processes is consistent with previous research describing alterations in mitochondrial activities and OXPHOS in GBM. Mitochondrial function has been associated with GBM progression and tumour-cell survival, while differences in oxidative phosphorylation have also been reported between molecular subtypes of GBM [16], [17]. The present findings therefore agree with previous evidence that mitochondrial metabolic processes are an important component of the molecular characteristics of GBM. However, a more significant enrichment result does not necessarily confirm that the biological processes are more or less active, and must be confirmed via other analyses.

Further inspection of the expression profiles of the individual respiratory complexes showed that genes belonging to Complexes I–V were predominantly down-regulated. This provides further detail in addition to the GO enrichment results, showing that the biological processes represented are accompanied by a generally negative direction of expression change among the genes involved. This may imply reduced mitochondrial activity, potentially leading to improper regulation of cell death-related functions, agreeing with what was proposed in previous literature in Section 2.4.3.

Additionally, the expression patterns of the apoptosis-associated genes also showed a relatively balanced distribution of upregulated and downregulated genes. This indicates that the transcriptional alterations affecting apoptosis-associated genes were not predominantly directed in a single direction, but instead involved changes in both upregulation and downregulation.

However, these interpretations from the computations remain speculative, as increase or reduction in mitochondrial respiratory and apoptotic activity cannot be determined from transcriptomic data alone [52], and experimental confirmation is still required.

5.2 ESTABLISHED APOPTOSIS–MITOCHONDRIAL RESPIRATORY RELATIONSHIP IN CLASSICAL GBM

The identification of CYCS within the relationship between apoptotic genes and mitochondrial respiratory complexes is consistent with past established biological connection [13], [18]. Rather than representing a newly identified mechanism, the result demonstrates that the computational analysis was able to capture a previously established link between mitochondrial respiratory function and apoptosis. More importantly, many previous computational studies of GBM have relied on TCGA-derived transcriptomic data, whereas the present analysis uses the independent CPTAC cohort, showing that this association is also computationally observed in a separate GBM dataset.

Furthermore, the present analysis examines this relationship specifically within Classical GBM, rather than considering GBM as a single molecular group. Thus, while the relationship between mitochondrial respiration, CYCS, and apoptosis itself is well established, its computational observation within Classical GBM using the CPTAC cohort provides a subtype-specific characterization of this relationship. This also provides a basis for investigating whether similar associations are observed in the remaining two major GBM subtypes in future studies.

5.3 ADDITIONAL NETWORK ASSOCIATIONS BETWEEN APOPTOSIS AND RESPIRATORY COMPLEXES

The number of associations varied between the respiratory complexes, with Complex I showing the largest number of connections, followed by Complex II, Complex IV, and Complex III, while no associations were identified for Complex V. This variation suggests that the relationship between apoptosis and mitochondrial respiratory complexes may extend beyond the well-established role of CYCS and involve multiple components of the respiratory chain.

The high number of associations identified for Complex I may be related to the multiple established connections between this complex and apoptotic regulation. In addition to its central role in electron transport, Complex I is a major site of mitochondrial ROS production, while several of its subunits have been directly associated with apoptotic processes. For example, NDUFA13/GRIM-19 has been identified as a regulator of cell death, while NDUFS1 can be targeted by caspases during apoptosis [53], [54]. The relatively high connectivity of Complex I in the present network may therefore reflect the diverse biological links between Complex I and apoptosis-associated processes.

Complex II showed the second-highest number of associations in the network. This finding is consistent with previous evidence describing Complex II as an important mediator of cell-death signalling. Unlike the other respiratory complexes, Complex II also directly links the tricarboxylic acid cycle with the electron transport chain, allowing metabolic changes to influence respiratory and apoptotic signalling. In particular, Complex II has been proposed to function as a general sensor of apoptotic stimuli, with alterations in its activity capable of promoting ROS generation and cell death [55].

Complex IV showed the third-highest number of associations in the network, with 10 connections. As the terminal complex of the electron transport chain, Complex IV receives electrons from cytochrome c and transfers them to molecular oxygen, completing the electron transfer process [15]. Its direct interaction with cytochrome c provides an established connection to the apoptotic pathway, as cytochrome c can also function as a mediator of intrinsic apoptosis following its release from mitochondria [56]. The relatively high number of associations identified for Complex IV may therefore reflect its position at the intersection of mitochondrial respiration and the CYCS-associated apoptotic pathway.

Complex III transfers electrons to cytochrome c, which subsequently transfers electrons to Complex IV as part of the respiratory chain [15]. Its primarily intermediate role in electron transfer may contribute to its lower connection count compared with the aforementioned two complexes. Furthermore, respiratory activity involving Complex III has been experimentally associated with cytochrome c release during apoptosis [57]. Although Complex III showed fewer associations than Complexes I and II, these findings indicate that its connection to apoptosis remains relevant within the network.

In contrast, no associations were identified for Complex V. Complex V differs functionally from Complexes I–IV because it does not directly participate in electron transfer or proton pumping; instead, ATP synthase uses the proton-motive force generated by the respiratory chain to produce ATP [15]. Therefore, it suggests that the network captures interactions that are more direct and not as passive as the Complex V-apoptosis interaction.

It should be emphasized that the studies discussed above do not establish the observed order of connectivity as an absolute biological hierarchy. Rather, they provide possible biological interpretations for the differences in connection counts observed in the present network, while the network itself provides the basis for identifying these associations in Classical GBM.

5.4 LIMITATIONS AND FUTURE WORK

The findings align closely with previous research results. However, the underlying transcriptomic data used to compare the GBM samples and GTEx controls are not guar-

anteed to be entirely technically and biologically suitable for a one-to-one comparison, although a justification for the datasets used was presented in Section 3.1. There is also a concern regarding the number of GTEx control samples, which was limited to only 7, as transcriptomic data from the remaining 3 samples were inaccessible. A larger number of control samples could provide greater confidence in the observed differences between the GBM and control groups.

A main concern of the methodology is the trade-off involved in selecting the confidence threshold for the identified STRING interactions. Opting for a higher confidence threshold increases the likelihood that the identified interactions are well-supported, but may eliminate potential interactions that are less well documented. Conversely, using a lower confidence threshold may provide a more comprehensive representation of the system, but can introduce additional noise and make the resulting network less clear. A possible direction for future work would therefore be to compare networks generated using different confidence thresholds and investigate how the identified associations change.

Regarding the networks, the analysis in this thesis was limited to counting the edges representing known associations. While the number of connections provides an overview of the connectivity of each respiratory complex, this metric alone provides limited insight beyond identifying whether associations exist between entities. If the work were to be extended further, additional network metrics in combination with biologically relevant measures could provide greater insight into the structure and organization of the biological systems. For example, measures of centrality or network clustering could be used to identify genes or groups of genes that occupy important positions within the networks.

As discussed in Section 5, the analysis in this thesis only covered the Classical GBM subtype. Extending the same analysis to the remaining Proneural and Mesenchymal subtypes, and potentially to other forms of programmed cell death, could provide a broader view of the interactions underlying GBM's proliferative behaviour.

Additionally, as the entities investigated ultimately depend on proteins to carry out their biological functions, integrating proteomic data with the present transcriptomic analysis could provide a more robust interpretation of the results and extend the work into a multi-omics analysis. Such an approach could reveal differences between transcriptomic changes and their corresponding protein products, as gene expression does not necessarily translate directly into protein abundance or function.

As discussed throughout Section 5, the computational results obtained in this thesis remain speculative in terms of their functional interpretation and require further experimental confirmation. This is consistent with the nature of bioinformatics research discussed in Section 2.5.1, where computational findings can provide hypotheses and identify potential biological relationships that subsequently require experimental validation. Overall, the analysis could also be strengthened by incorporating additional

statistical analyses to provide further quantitative support for the observed relationships, rather than relying primarily on comparisons with previous literature.

6 CONCLUSIONS

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